

MOHAMMAD MEHDI ASGARI

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Education

2018-2022	M.Sc. of Seafood Processing / Sari Agricultural Sciences and Natural Resources University GPA: 16.48/20.00
2014-2018	B.Sc. of Natural Science Engineering - Fisheries / Sari Agricultural Sciences and Natural Resources University Majoring in Aquaculture Breeding and Culture

Research Experience

University of Tehran 2025-2026	Ultrasound Thawing Enhances Myofibrillar Protein Integrity and Quality Attributes of Common Carp (<i>Cyprinus carpio</i>) Fillets https://doi.org/10.1002/fsn3.71833 - Applied three thawing methods (Ultrasound, Water Immersion, Microwave) - Analyzed effects on fillet and protein structure using various analytical techniques (such as SEM, TPA, Color, DLS, Zeta Potential, SDS-Page). Result: The Ultrasound-assisted Thawing (UT) method demonstrated a 49% better performance in minimizing myofibrillar protein aggregation compared to the worst-performing method (Water Immersion Thawing). Furthermore, UT significantly enhanced water-holding capacity, resulting in 10% less cooking loss (18.95%) compared to the highest loss observed in WT (21.01%) . These results, along with the highest Zeta potential observed in the UT group , establish ultrasound thawing as the optimal protocol for preserving protein integrity and functionality
University of Tehran 2024 - 2025	Studying the Combined Effect of Sous-vide and Shirazi Thyme Extract (<i>Zataria multiflora</i>) on the Quality and Shelf Life of Western White Shrimp (<i>Litopenaeus vannamei</i>) / Advisor & Experimental Designer Wide range of experiment done in each section: - Microbiological examination such as Enterobacteriaceae & Total Bacterial Load - Chemical Parameters such as TVB-N & TBA - Physical Parameters such as TPA & Color & Sensory (smell, Color, acceptance) Result: Evaluated sous-vide processing combined with 1% Shirazi thyme extract for Pacific white shrimp (<i>Litopenaeus vannamei</i>) preservation, extending refrigerated shelf life to 16 days while improving microbial stability, lipid oxidation control, and sensory quality.
University of Tehran 2024 - 2025	Investigation of the Feasibility of Producing Hydrolyzed Collagen from Nile Tilapia (<i>Oreochromis niloticus</i>) Skin Using a Microbial Method / Advisor & Experimental Designer - Extracted collagen from Nile tilapia skin and hydrolysed it using Trichoderma ghanense (DH $\approx$ 27%). HPLC and Structural analyses (FT-IR, DSC) confirmed peptide formation and enhanced thermal stability. Result: Developed hydrolyzed collagen from Nile tilapia skin through microbial fermentation, resulting in low-molecular-weight peptides, improved solubility, and potential functional applications.

Research Interest

Food Processing & Preservation / Protein Structure and Functionality / Protein Aggregation/ Food Texture and Rheology/ Bioactive Lipids & Functional Ingredients / Enzymatic Hydrolysis/ Food Quality and Safety/ Ultrasound- and Microwave-Assisted Processing/ Sustainable Food Packaging

Academic Project

Oct 2025-Current	Applied Quality & Processing-Relevant Assessment of Rainbow Trout & Brown Trout (Tar & Lar Lakes) Conducted a multidisciplinary field study on <i>O. mykiss</i> and <i>S. trutta</i> assessing population traits, morphology, feeding patterns and water-quality indicators, with emphasis on how environmental conditions influence muscle quality, biochemical profiles , and processing-relevant attributes that affect shelf-life and nutritional value in aquaculture products .
Nov 2024- Nov 2025	Functional Quality Evaluation of Pike (<i>Esox lucius</i>) for Processing Applications (Ovan Lake) Led an ecological and physiological assessment of pike populations, integrating behaviour, habitat parameters, and diet composition with measurements linked to fillet texture, fat-protein balance , and raw-material suitability for value-added processing , supporting data-driven approaches to product development.
Oct 2023 - Sep 2024	Quality-Focused Eco-Physiological Study of Brown Trout (Chalus River) Investigated population characteristics, habitat conditions, feeding ecology, and health indicators of <i>S. trutta</i> , with additional focus on muscle integrity, biochemical stability , and environmental impacts on post-harvest quality , providing insights relevant to processing performance and functional food applications.

WORK
Experience

University of Tehran
Oct 2023 - Current

SnappFood
June 2020 - May 2023

Nilofar Abi Co.
Feb 2018 - May 2020

Behbudi Aqua
Jun2015-Jan2018

Teacher
Assistant

University of Tehran
Oct 2023 - Current

University of Tehran
Oct 2023 - Current

University of Tehran
Oct 2023 - Current

Skills

Selected
Conferences

Nov 2025

Nov 2024

Honours&
Awards

Language

Research and Lab Assistant

- Led 10+ field visits to farms and seafood plants; boosted student engagement by 40%.
- Taught 3 courses in quality control and aquaculture to 80+ students; >90% satisfaction.
- Supervised 4+ M.Sc. and PhD thesis in aquaculture technology and seafood quality.
- Established a reproduction and feeding-trial facility; increased research capacity by 35%.
- Supported lab operations (fisheries and seafood), equipment upkeep, and academic coordination.

Food Quality Controller

- Real-time issue resolution (200+ orders/day)
- Quality & hygiene audits (95% compliance)
- Onboarding & evaluating 20+ restaurants
- Cross-team coordination (restaurants-
logistics-support)
- Complaint rate reduction (~25%)
- Food quality control in digital services
- Customer experience monitoring
- Operational performance assessment
- Strong analytical & communication skills

Fish Farm Supervisor

- Managed daily operations across four cold-water ponds producing 30+ tons of salmonids.
- Led a team of 10 technicians, improving operational efficiency by ~25%.
- Monitored water quality, feeding schedules, and health status, keeping survival rates above 90%.
- Reduced early-life mortality of salmonids to below 11%.
- Improved fillet quality (texture, colour, nutritional profile) by an average of 6% annually.
- Supported sales planning and client relations, contributing to a 15% increase in customer retention.
- Gained hands-on expertise in broodstock management, aquaculture production cycles, and post-harvest quality control

Aquaculture Technician → Head of Fish Breeder and Holding Facility

- Promoted from technician to department head based on performance and leadership
- Designed 100+ saltwater RAS aquarium systems, improving efficiency ~20%
- Monitored coral and fish sections; maintained water-quality parameters within ±3% of optimal levels
- Supervised broodstock health of 150+ marine species, improving hatch success by 15% annually
- Diagnosed and treated 30+ disease cases, reducing mortality by 25%
- Oversaw quarantine operations for 100+ monthly imports with zero cross-contamination
- Delivered 12+ staff training sessions annually, improving team performance by 18%

“Quality Control of Fishery Products” – Delivered full lectures and practical sessions to undergraduate students

“Cold-Water Fish Breeding” – Delivered full lectures and practical sessions to undergraduate students

“Breeding and Rearing of Ornamental Fish” – Delivered full lectures and practical sessions to undergraduate students

Hard Skills

Laboratory Techniques & Experimental Design, PCR, HPLC, Microbial Culture, SEM, Python, SPSS, HTML, Written & Verbal Communication Skills, UV-VIS- SDS-PAGE, AFM, TPA

Soft Skills

Communication & Presentation, Leadership & Teaching, Adaptability & Time Management, Teamwork & Collaboration, Problem Solving & Critical Thinking

- Ninth International Conference on Interdisciplinary Studies in Food industry and Nutrition Science of Iran
- Eighth International Conference on Interdisciplinary Studies in Food industry and Nutrition Science of Iran
- 10th International Conference on Food Science, Organic Agriculture and Food Security

- Ranked Top %1 Among Participants in The Undergraduate Nationwide Entrance Exam
- Ranked 114th Among +4500 Participants in Master Nationwide Entrance Exam
- Ranked 1st Among Fisheries Engineering, Seafood Processing Students

Language: Persian – Native / English – IELTS 7.5